# Evaluation runs: **replog** (replicated-log-aneris)

generated 2026-11-08 from brain-data 34d8e8fae 2026-11-08 19:23:20 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". $G_{ref}$, $G_{cand}$ = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is $G_{ref}^{expl}$); $G_x$ = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

## stateless

### stateless: pair **asyn-fault**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 36 | conforms | 0.0 | 4 | 1 h, 250 GB | 3 | 9 | brain03, 2026-11-02 |
| 3 | 20 | 400 | conforms | 0.1 | 4 | 1 h, 250 GB | 6 | 41 | brain03, 2026-11-02 |
| 4 | 70 | 4900 | conforms | 0.2 | 4 | 1 h, 250 GB | 11 | 230 | brain03, 2026-11-02 |
| 5 | 252 | 63504 | conforms | 0.8 | 4 | 1 h, 250 GB | 24 | 1416 | brain03, 2026-11-02 |
| 6 | 924 | 853776 | conforms | 8.4 | 10 | 1 h, 250 GB | 55 | 12007 | brain03, 2026-11-02 |
| 7 | 3432 | 11778624 | conforms | 120.9 | 78 | 1 h, 250 GB | 264 | 278608 | brain03, 2026-11-02 |
| 8 | 12870 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 30 | conforms | 0.0 | 4 | 1 h, 250 GB | 3 | 14 | brain03, 2026-11-02 |
| 3 | 120 | 280 | conforms | 0.1 | 4 | 1 h, 250 GB | 7 | 182 | brain03, 2026-11-02 |
| 4 | 1680 | 2940 | conforms | 0.5 | 4 | 1 h, 250 GB | 22 | 8776 | brain03, 2026-11-02 |
| 5 | 30240 | 33264 | conforms | 12.5 | 4 | 1 h, 250 GB | 85 | 651755 | brain03, 2026-11-02 |
| 6 | 665280 | 396396 | conforms | 404.9 | 6 | 1 h, 250 GB | 287 | 34200952 | brain03, 2026-11-02 |
| 7 | 17297280 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 30 | conforms | 0.0 | 4 | 1 h, 250 GB | 3 | 9 | brain03, 2026-11-02 |
| 3 | 20 | 280 | conforms | 0.1 | 4 | 1 h, 250 GB | 6 | 41 | brain03, 2026-11-02 |
| 4 | 70 | 2940 | conforms | 0.2 | 4 | 1 h, 250 GB | 13 | 275 | brain03, 2026-11-02 |
| 5 | 252 | 33264 | conforms | 0.8 | 4 | 1 h, 250 GB | 26 | 1629 | brain03, 2026-11-02 |
| 6 | 924 | 396396 | conforms | 4.2 | 6 | 1 h, 250 GB | 85 | 18571 | brain03, 2026-11-02 |
| 7 | 3432 | 4907760 | conforms | 47.1 | 18 | 1 h, 250 GB | 148 | 99875 | brain03, 2026-11-02 |
| 8 | 12870 | 62573940 | conforms | 774.9 | 286 | 1 h, 250 GB | 323 | 771832 | brain03, 2026-11-02 |
| 9 | 48620 | | 🕓 (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

### stateless: pair **asyn-fault**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | sweeps | $G_{ref}^{expl}$ | run |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 144 | 12 | violation | 0.1 | 4 | 1 h, 250 GB | 16 | 161 | brain03, 2026-11-02 |
| 3 | 120 | 14400 | 2880 | violation | 6.0 | 4 | 1 h, 250 GB | 2889 | 345834 | brain03, 2026-11-02 |
| 4 | 1680 | | | (1 h) | | | 1 h, 250 GB | | | brain03, 2026-11-02 |

**Not run (stateless).** planned but never scheduled (grid points beyond the ladder): 7 correct bag, 8 asyn-fault fifo, 8 correct fifo, 8 fifo correct, 8 fifo asyn-fault, 9 fifo correct, 9 fifo asyn-fault, 10 fifo correct, 10 fifo asyn-fault.

skipped: a smaller size had reached its limit: 5 asyn-fault bag, 6 asyn-fault bag, 7 asyn-fault bag, 8 correct bag, 8 asyn-fault bag, 9 asyn-fault bag, 9 asyn-fault fifo, 9 correct bag, 10 correct fifo, 10 correct bag, 10 asyn-fault fifo.

# stateful

## stateful: pair **asyn-fault**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 6 | 36 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 20 | 400 | conforms | 0.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 70 | 4900 | conforms | 0.2 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5 | 252 | 63504 | conforms | 0.7 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6 | 924 | 853776 | conforms | 7.4 | 10 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7 | 3432 | 11778624 | conforms | 124.8 | 64 | 48 h, 250 GB | brain03, 2026-11-06 |
| 8 | 12870 | 165636900 | conforms | 2557.7 | 1068 | 48 h, 250 GB | brain05, 2026-11-02 |
| 9 | 48620 | 2363904400 | conforms | 41296.0 | 14812 | 48 h, 250 GB | brain05, 2026-11-02 |
| 10 | | | (48 h) | | | 48 h, 250 GB | brain01, 2026-11-05 |
| 11 | | | (48 h) | | | 48 h, 250 GB | brain04, 2026-11-03 |
| 14 | | | (48 h) | | | 48 h, 200 GB | volta11, 2026-11-05 |

## stateful: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 12 | 30 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 120 | 280 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 1680 | 2940 | conforms | 0.3 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5 | 30240 | 33264 | conforms | 1.1 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6 | 665280 | 396396 | conforms | 9.4 | 6 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7 | 17297280 | 4907760 | conforms | 200.0 | 58 | 48 h, 250 GB | brain03, 2026-11-06 |
| 8 | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-02 |

## stateful: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|
| 2 | 6 | 30 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 3 | 20 | 280 | conforms | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 4 | 70 | 2940 | conforms | 0.2 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 5 | 252 | 33264 | conforms | 0.8 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 6 | 924 | 396396 | conforms | 4.3 | 4 | 1 h, 250 GB | brain02, 2026-11-02 |
| 7 | 3432 | 4907760 | conforms | 50.3 | 34 | 48 h, 250 GB | brain03, 2026-11-06 |
| 8 | 12870 | 62573940 | conforms | 894.4 | 290 | 48 h, 250 GB | brain05, 2026-11-02 |
| 9 | 48620 | 816621520 | conforms | 14594.0 | 3194 | 48 h, 250 GB | brain05, 2026-11-02 |
| 10 | | | (48 h) | | | 48 h, 250 GB | brain04, 2026-11-02 |

## stateful: pair **asyn-fault**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 144 | 12 | violation | 0.0 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 3 | 120 | 14400 | 2880 | violation | 0.5 | 4 | 1 h, 250 GB | brain02, 2026-11-01 |
| 4 | 1680 | 2822400 | 912240 | violation | 19.0 | 10 | 1 h, 250 GB | brain02, 2026-11-01 |

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | run |
|---|---|---|---|---|---|---|---|---|
| 5 | | | | (1 h) | | | 1 h, 250 GB | brain02, 2026-11-01 |

**Not run (stateful).** planned but never scheduled (grid points beyond the ladder): 5 asyn-fault bag, 9 asyn-fault fifo, 11 correct fifo, 11 asyn-fault fifo, 12 fifo correct, 12 correct fifo, 12 asyn-fault fifo, 13 correct fifo, 13 asyn-fault fifo, 14 correct fifo, 15 correct fifo, 15 asyn-fault fifo, 16 asyn-fault fifo, 16 correct fifo.

skipped: a smaller size had reached its limit: 6 asyn-fault bag, 7 asyn-fault bag, 8 asyn-fault bag, 9 asyn-fault bag, 9 correct bag, 10 correct fifo, 10 correct bag, 10 asyn-fault fifo.

left out by the table rules (ok): 7 correct fifo, 7 asyn-fault fifo, 7 correct bag, 7 asyn-fault fifo.x3.t16.c48h.r7, 7 correct bag.x3.t16.c48h.r7, 7 correct fifo.x3.t16.c48h.r7, 8 asyn-fault fifo, 8 correct fifo.

left out by the table rules (timeout): 9 correct fifo, 9 asyn-fault fifo.

left out by the table rules (invalid: foreign load): 11 fifo correct, 12 fifo asyn-fault.

# offline

## offline: pair **asyn-fault**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 36 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 20 | 400 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 70 | 4900 | conforms | 0.3 | 8 | 10 min, 16 GB | 0.0 | 0.3 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 252 | 63504 | conforms | 1.1 | 8 | 10 min, 16 GB | 0.1 | 1.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 6 | 924 | 853776 | conforms | 8.6 | 20 | 10 min, 16 GB | 0.1 | 8.5 | 0.1 | in-memory | brain05, 2026-11-02 |
| 7 | 3432 | 11778624 | conforms | 145.2 | 108 | 48 h, 250 GB | 0.3 | 144.8 | 0.4 | merge | brain03, 2026-11-06 |
| 8 | 12870 | 165636900 | conforms | 2958.3 | 1144 | 1 h, 250 GB | 0.8 | 2956.4 | 1.9 | merge | brain05, 2026-11-03 |

## offline: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 30 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.0 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 120 | 280 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.1 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 1680 | 2940 | conforms | 0.4 | 8 | 10 min, 16 GB | 0.1 | 0.4 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 30240 | 33264 | conforms | 1.2 | 32 | 10 min, 16 GB | 0.7 | 1.1 | 0.2 | in-memory | brain05, 2026-11-02 |
| 6 | 665280 | 396396 | conforms | 9.4 | 74 | 1 h, 250 GB | 7.8 | 5.2 | 1.6 | merge | brain05, 2026-11-03 |
| 7 | 17297280 | 4907760 | conforms | 192.3 | 642 | 48 h, 250 GB | 183.7 | 60.5 | 8.6 | merge | brain03, 2026-11-06 |

## offline: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 30 | conforms | 0.1 | 8 | 10 min, 16 GB | 0.0 | 0.1 | 0.0 | in-memory | brain05, 2026-11-02 |
| 3 | 20 | 280 | conforms | 0.2 | 8 | 10 min, 16 GB | 0.0 | 0.2 | 0.0 | in-memory | brain05, 2026-11-02 |
| 4 | 70 | 2940 | conforms | 0.3 | 8 | 10 min, 16 GB | 0.0 | 0.3 | 0.0 | in-memory | brain05, 2026-11-02 |
| 5 | 252 | 33264 | conforms | 1.1 | 28 | 10 min, 16 GB | 0.1 | 1.0 | 0.1 | in-memory | brain05, 2026-11-02 |
| 6 | 924 | 396396 | conforms | 7.3 | 72 | 1 h, 250 GB | 0.1 | 5.7 | 1.6 | merge | brain05, 2026-11-03 |
| 7 | 3432 | 4907760 | conforms | 69.2 | 559 | 48 h, 250 GB | 0.3 | 60.8 | 8.3 | merge | brain03, 2026-11-06 |
| 8 | 12870 | 62573940 | conforms | 1292.2 | 4588 | 1 h, 250 GB | 0.8 | 1195.8 | 96.4 | merge | brain05, 2026-11-03 |

## offline: pair **asyn-fault**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 144 | 1 | violation | 0.1 | 8 | 1 h, 250 GB | 0.1 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 120 | 14400 | 10 | violation | 0.6 | 8 | 1 h, 250 GB | 0.1 | 0.5 | 0.1 | merge | brain05, 2026-11-03 |
| 4 | 1680 | 2822400 | 74 | violation | 18.6 | 26 | 1 h, 250 GB | 0.1 | 18.4 | 0.3 | merge | brain05, 2026-11-03 |

**Not run (offline).** planned but never scheduled (grid points beyond the ladder): 5 asyn-fault bag, 8 asyn-fault fifo, 8 fifo correct, 8 fifo asyn-fault, 9 asyn-fault fifo, 9 fifo correct, 9 correct bag, 9 correct fifo, 9 fifo asyn-fault, 10 fifo asyn-fault.

skipped: a smaller size had reached its limit: 5 asyn-fault bag, 6 asyn-fault bag, 7 asyn-fault bag, 8 asyn-fault bag, 9 asyn-fault bag, 9 asyn-fault fifo, 9 correct bag, 9 correct fifo, 10 correct fifo, 10 fifo correct, 10 correct bag, 10 asyn-fault fifo.

left out by the table rules (ok): 7 correct fifo, 7 asyn-fault fifo, 7 correct bag, 7 asyn-fault fifo.x3.m.c48h.r7, 7 correct bag.x3.m.c48h.r7, 7 correct fifo.x3.m.c48h.r7.

refused (an input missing at run time): 8 correct bag.

# cross offline, Java implementation against the Rust specification

## cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 30 | conforms | 6.4 | 169 | 4 h, 250 GB | 0.1 | 6.4 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 120 | 280 | conforms | 9.2 | 166 | 10 min, 250 GB | 0.1 | 9.1 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 1680 | 2940 | conforms | 21.6 | 3071 | 10 min, 250 GB | 0.1 | 21.6 | 0.1 | merge | brain01, 2026-11-03 |

## cross offline, Java implementation against the Rust specification: pair **asyn-fault**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 144 | 1 | violation | 4.7 | 163 | 10 min, 250 GB | 0.1 | 4.7 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 120 | 14400 | 10 | violation | 25.4 | 4412 | 10 min, 250 GB | 0.1 | 25.1 | 0.3 | merge | brain01, 2026-11-03 |

**Not run (cross offline, Java implementation against the Rust specification).** skipped: a smaller size had reached its limit: 4 bag asyn-fault, 10 fifo correct.

planned but never scheduled (grid points beyond the ladder): 5 bag correct, 8 fifo correct, 8 fifo asyn-fault, 9 fifo correct, 9 fifo asyn-fault, 10 fifo asyn-fault.

refused (an input missing at run time): 2 fifo correct, 2 fifo asyn-fault, 3 fifo asyn-fault, 3 fifo correct, 4 fifo asyn-fault, 4 fifo correct.

# cross offline, Rust implementation against the Java specification

## cross offline, Rust implementation against the Java specification: pair **asyn-fault**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 36 | conforms | 4.3 | 162 | 10 min, 250 GB | 4.1 | 0.1 | 0.2 | merge | brain01, 2026-11-03 |
| 3 | 20 | 400 | conforms | 8.1 | 155 | 10 min, 250 GB | 8.0 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 70 | 4900 | conforms | 8.2 | 151 | 10 min, 250 GB | 8.2 | 0.3 | 0.0 | merge | brain01, 2026-11-03 |

## cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 30 | conforms | 6.5 | 152 | 4 h, 250 GB | 6.5 | 0.0 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 120 | 280 | conforms | 4.5 | 161 | 10 min, 250 GB | 4.5 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 1680 | 2940 | conforms | 8.4 |  | 10 min, 250 GB | 8.4 | 0.4 | 0.1 | merge | brain01, 2026-11-03 |

## cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 30 | conforms | 6.1 | 160 | 4 h, 250 GB | 6.1 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 20 | 280 | conforms | 8.3 | 155 | 10 min, 250 GB | 8.0 | 0.2 | 0.3 | merge | brain01, 2026-11-03 |
| 4 | 70 | 2940 | conforms | 8.3 | 151 | 10 min, 250 GB | 8.2 | 0.3 | 0.2 | merge | brain01, 2026-11-03 |

### cross offline, Rust implementation against the Java specification: pair **asyn-fault**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 144 | 1 | violation | 3.9 | 153 | 10 min, 250 GB | 3.9 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 120 | 14400 | 10 | violation | 4.6 | 161 | 10 min, 250 GB | 4.5 | 0.5 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 1680 | 2822400 | 74 | violation | 26.3 | 271 | 48 h, 250 GB | 6.1 | 26.3 | 0.0 | merge | brain15, 2026-11-07 |

**Not run (cross offline, Rust implementation against the Java specification).** skipped: a smaller size had reached its limit: 10 fifo asyn-fault, 10 fifo correct.

planned but never scheduled (grid points beyond the ladder): 5 fifo asyn-fault, 5 fifo correct, 5 bag correct, 8 fifo correct, 8 fifo asyn-fault, 9 fifo correct, 9 fifo asyn-fault.

## cross online, Java implementation (JMC) against the Rust specification export

### cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 10 | 184756 | | (48 h) | | | 48 h, 200 GB | | volta10, 2026-11-05 |

**Not run (cross online, Java implementation (JMC) against the Rust specification export).** planned but never scheduled (grid points beyond the ladder): 2 fifo correct, 2 fifo asyn-fault, 2 bag correct, 2 bag asyn-fault, 3 bag asyn-fault, 3 bag correct, 3 fifo asyn-fault, 3 fifo correct, 4 bag correct, 4 fifo asyn-fault, 4 fifo correct, 8 fifo correct, 8 fifo asyn-fault, 9 fifo correct, 9 fifo asyn-fault, 10 fifo asyn-fault.

## cross online, Rust implementation (TraceForge) against the Java specification export

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **asyn-fault**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 36 | conforms | 3.2 | 160 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 20 | 400 | conforms | 3.3 | 161 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 70 | 4900 | conforms | 3.8 | 162 | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 30 | conforms | 3.1 | 160 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 120 | 280 | conforms | 3.6 | 167 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 1680 | 2940 | conforms | 7.7 | 583 | 1 h, 250 GB | | brain01, 2026-11-02 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 6 | 30 | conforms | 3.2 | 160 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 20 | 280 | conforms | 3.3 | 161 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 70 | 2940 | conforms | 3.8 | 162 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 10 | 184756 | | oom (80 GB) | | | 48 h, 80 GB | | pinky02, 2026-11-05 |

### cross online, Rust implementation (TraceForge) against the Java specification export: pair **asyn-fault**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 12 | 144 | 12 | violation | 3.1 | 160 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 3 | 120 | 14400 | 2880 | violation | 4.1 | 167 | 1 h, 250 GB | 4 | brain01, 2026-11-02 |

**Not run (cross online, Rust implementation (TraceForge) against the Java specification export).** planned but never scheduled (grid points beyond the ladder): 8 fifo correct, 8 fifo asyn-fault, 9 fifo correct, 9 fifo asyn-fault, 10 fifo asyn-fault.

## twins offline, Rust implementation against its Java twin

### twins offline, Rust implementation against its Java twin: pair **asyn-fault**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 144 | 144 | conforms | 5.2 | 163 | 10 min, 250 GB | 4.7 | 0.1 | 0.6 | merge | brain01, 2026-11-03 |
| 3 | 14400 | 14400 | conforms | 25.2 | 4412 | 10 min, 250 GB | 25.1 | 0.5 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **asyn-fault**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 36 | 36 | conforms | 4.1 | 160 | 10 min, 250 GB | 4.1 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 400 | 400 | conforms | 9.6 | 160 | 10 min, 250 GB | 9.6 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 4900 | 4900 | conforms | 15.1 | 3371 | 10 min, 250 GB | 14.9 | 0.3 | 0.2 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 6.4 | 169 | 4 h, 250 GB | 6.4 | 0.0 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 280 | 280 | conforms | 9.1 | 166 | 10 min, 250 GB | 9.1 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 2940 | 2940 | conforms | 21.9 | 3071 | 10 min, 250 GB | 21.6 | 0.4 | 0.3 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 6.9 | 165 | 4 h, 250 GB | 6.9 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 280 | 280 | conforms | 5.4 | 182 | 10 min, 250 GB | 5.3 | 0.2 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 2940 | 2940 | conforms | 21.8 | 3022 | 10 min, 250 GB | 21.6 | 0.3 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **asyn-fault**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 2940 | 2822400 | 97 | violation | 26.3 | 420 | 48 h, 250 GB | 10.0 | 26.3 | 0.0 | merge | brain15, 2026-11-07 |

### twins offline, Rust implementation against its Java twin: pair **asyn-fault**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 144 | 1 | violation | 7.4 | 166 | 10 min, 250 GB | 7.1 | 0.1 | 0.3 | merge | brain01, 2026-11-03 |
| 3 | 280 | 14400 | 12 | violation | 9.1 | 166 | 10 min, 250 GB | 9.1 | 0.5 | 0.0 | merge | brain01, 2026-11-03 |

### twins offline, Rust implementation against its Java twin: pair **asyn-fault**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 36 | | conforms* | 7.3 | 153 | 10 min, 250 GB | 7.3 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 280 | 400 | 2 | violation* | 5.5 | 182 | 10 min, 250 GB | 5.3 | 0.1 | 0.2 | merge | brain01, 2026-11-03 |
| 4 | 2940 | 4900 | 23 | violation* | 22.6 | 3022 | 10 min, 250 GB | 21.6 | 0.3 | 1.0 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Rust implementation against its Java twin).** skipped: a smaller size had reached its limit: 4 bag asyn-fault.

planned but never scheduled (grid points beyond the ladder): 5 fifo asyn-fault, 5 fifo correct, 5 bag correct, 8 fifo correct, 8 fifo asyn-fault, 9 fifo correct, 9 fifo asyn-fault.

## twins offline, Java implementation against its Rust twin

### twins offline, Java implementation against its Rust twin: pair **asyn-fault**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 144 | 144 | conforms | 4.7 | 163 | 10 min, 250 GB | 0.1 | 4.7 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 14400 | 14400 | conforms | 25.3 | 4412 | 10 min, 250 GB | 0.5 | 25.1 | 0.2 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **asyn-fault**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 36 | 36 | conforms | 4.1 | 160 | 10 min, 250 GB | 0.1 | 4.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 400 | 400 | conforms | 9.6 | 160 | 10 min, 250 GB | 0.1 | 9.6 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 4900 | 4900 | conforms | 14.9 | 3371 | 10 min, 250 GB | 0.3 | 14.9 | 0.0 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 6.4 | 169 | 4 h, 250 GB | 0.0 | 6.4 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 280 | 280 | conforms | 9.2 | 166 | 10 min, 250 GB | 0.1 | 9.1 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 2940 | 2940 | conforms | 21.6 | 3071 | 10 min, 250 GB | 0.4 | 21.6 | 0.1 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 6.9 | 165 | 4 h, 250 GB | 0.1 | 6.9 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 280 | 280 | conforms | 5.4 | 182 | 10 min, 250 GB | 0.2 | 5.3 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 2940 | 2940 | conforms | 21.9 | 3022 | 10 min, 250 GB | 0.3 | 21.6 | 0.2 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **asyn-fault**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 144 | 1 | violation | 4.8 | 163 | 10 min, 250 GB | 0.0 | 4.7 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 280 | 14400 | 12 | violation | 25.1 | 4412 | 10 min, 250 GB | 0.1 | 25.1 | 0.0 | merge | brain01, 2026-11-03 |

### twins offline, Java implementation against its Rust twin: pair **asyn-fault**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 36 | | conforms* | 4.1 | 160 | 10 min, 250 GB | 0.1 | 4.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 280 | 400 | 2 | violation* | 9.6 | 160 | 10 min, 250 GB | 0.2 | 9.6 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 2940 | 4900 | 23 | violation* | 15.1 | 3371 | 10 min, 250 GB | 0.3 | 14.9 | 0.2 | merge | brain01, 2026-11-03 |

**Not run (twins offline, Java implementation against its Rust twin).** skipped: a smaller size had reached its limit: 4 bag asyn-fault.

planned but never scheduled (grid points beyond the ladder): 5 fifo asyn-fault, 5 fifo correct, 5 bag correct, 8 fifo correct, 8 fifo asyn-fault, 9 fifo correct, 9 fifo asyn-fault.

## context-free twins offline, Rust implementation against the context-free Java port

### context-free twins offline, Rust implementation against the context-free Java port: pair **asyn-fault**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 144 | 144 | conforms | 5.3 | 174 | 10 min, 250 GB | 5.3 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 14400 | 14400 | conforms | 39.1 | 4571 | 10 min, 250 GB | 39.0 | 0.5 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **asyn-fault**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 36 | 36 | conforms | 7.7 | 161 | 10 min, 250 GB | 7.7 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 400 | 400 | conforms | 6.7 | 184 | 10 min, 250 GB | 6.4 | 0.1 | 0.2 | merge | brain01, 2026-11-03 |
| 4 | 4900 | 4900 | conforms | 15.8 | 3401 | 10 min, 250 GB | 15.4 | 0.3 | 0.4 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 7.0 | 160 | 4 h, 250 GB | 7.0 | 0.0 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 280 | 280 | conforms | 5.6 | 170 | 10 min, 250 GB | 5.6 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 2940 | 2940 | conforms | 17.0 | 2291 | 10 min, 250 GB | 16.9 | 0.4 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair correct, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 6.8 | 162 | 4 h, 250 GB | 6.8 | 0.1 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 280 | 280 | conforms | 5.0 | 176 | 10 min, 250 GB | 4.9 | 0.2 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 2940 | 2940 | conforms | 15.3 | | 10 min, 250 GB | 15.3 | 0.3 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair asyn-fault, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 2940 | 2822400 | 97 | violation | 26.3 | 2507 | 48 h, 250 GB | 10.6 | 26.3 | 0.0 | merge | brain15, 2026-11-07 |

### context-free twins offline, Rust implementation against the context-free Java port: pair asyn-fault, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 144 | 1 | violation | 5.2 | 176 | 10 min, 250 GB | 5.1 | 0.1 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 280 | 14400 | 12 | violation | 5.7 | 169 | 10 min, 250 GB | 5.6 | 0.5 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Rust implementation against the context-free Java port: pair asyn-fault, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 36 | | conforms* | 4.4 | 165 | 10 min, 250 GB | 4.3 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 280 | 400 | 2 | violation* | 4.9 | 176 | 10 min, 250 GB | 4.9 | 0.1 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 2940 | 4900 | 23 | violation* | 15.5 | 2772 | 10 min, 250 GB | 15.3 | 0.3 | 0.2 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Rust implementation against the context-free Java port).** skipped: a smaller size had reached its limit: 4 bag asyn-fault.

planned but never scheduled (grid points beyond the ladder): 5 fifo asyn-fault, 5 fifo correct, 5 bag correct, 8 fifo correct, 8 fifo asyn-fault, 9 fifo correct, 9 fifo asyn-fault.

## context-free twins offline, Java port against the Rust implementation

### context-free twins offline, Java port against the Rust implementation: pair asyn-fault, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 144 | 144 | conforms | 5.3 | 174 | 10 min, 250 GB | 0.1 | 5.3 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 14400 | 14400 | conforms | 39.0 | 4571 | 10 min, 250 GB | 0.5 | 39.0 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair asyn-fault, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 36 | 36 | conforms | 7.7 | 161 | 10 min, 250 GB | 0.1 | 7.7 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 400 | 400 | conforms | 6.4 | 184 | 10 min, 250 GB | 0.1 | 6.4 | 0.0 | merge | brain01, 2026-11-03 |

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 4900 | 4900 | conforms | 15.5 | 3401 | 10 min, 250 GB | 0.3 | 15.4 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 7.0 | 160 | 4 h, 250 GB | 0.0 | 7.0 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 280 | 280 | conforms | 5.7 | 170 | 10 min, 250 GB | 0.1 | 5.6 | 0.1 | merge | brain01, 2026-11-03 |
| 4 | 2940 | 2940 | conforms | 17.0 | 2291 | 10 min, 250 GB | 0.4 | 16.9 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 6.8 | 162 | 4 h, 250 GB | 0.1 | 6.8 | 0.0 | merge | brain05, 2026-11-03 |
| 3 | 280 | 280 | conforms | 5.3 | 176 | 10 min, 250 GB | 0.2 | 4.9 | 0.4 | merge | brain01, 2026-11-03 |
| 4 | 2940 | 2940 | conforms | 15.3 | 2772 | 10 min, 250 GB | 0.3 | 15.3 | 0.0 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **asyn-fault**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 144 | 1 | violation | 5.3 | 174 | 10 min, 250 GB | 0.0 | 5.3 | 0.0 | merge | brain01, 2026-11-03 |
| 3 | 280 | 14400 | 12 | violation | 39.1 | 4572 | 10 min, 250 GB | 0.1 | 39.0 | 0.1 | merge | brain01, 2026-11-03 |

### context-free twins offline, Java port against the Rust implementation: pair **asyn-fault**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | export spec | export impl | join | join kind | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 36 | | conforms* | 7.8 | 161 | 10 min, 250 GB | 0.1 | 7.7 | 0.1 | merge | brain01, 2026-11-03 |
| 3 | 280 | 400 | 2 | violation* | 6.4 | 184 | 10 min, 250 GB | 0.2 | 6.4 | 0.0 | merge | brain01, 2026-11-03 |
| 4 | 2940 | 4900 | 23 | violation* | 15.5 | 3401 | 10 min, 250 GB | 0.3 | 15.4 | 0.1 | merge | brain01, 2026-11-03 |

**Not run (context-free twins offline, Java port against the Rust implementation).** skipped: a smaller size had reached its limit: 4 bag asyn-fault, 8 fifo asyn-fault.

planned but never scheduled (grid points beyond the ladder): 5 fifo asyn-fault, 5 fifo correct, 5 bag correct, 8 fifo correct, 9 fifo correct, 9 fifo asyn-fault.

## context-free twins online, Java port (JMC) against the Rust implementation's export

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **asyn-fault**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 144 | 144 | conforms | 6.5 | 172 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 3 | 14400 | 14400 | conforms | 35.4 | 4724 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **asyn-fault**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 36 | 36 | conforms | 6.4 | 165 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 3 | 400 | 400 | conforms | 8.9 | 176 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 4 | 4900 | 4900 | conforms | 23.0 | 4020 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 3.4 | 182 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 280 | 280 | conforms | 8.6 | 173 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 4 | 2940 | 2940 | conforms | 19.5 | 2674 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 3.4 | 156 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 280 | 280 | conforms | 8.8 | 170 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 4 | 2940 | 2940 | conforms | 20.4 | 2705 | 1 h, 250 GB | | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **asyn-fault**, model bag, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 144 | 12 | violation | 7.0 | 176 | 1 h, 250 GB | 4119 | brain01, 2026-11-03 |
| 3 | 280 | 14400 | 3438 | violation | 35.4 | 4575 | 1 h, 250 GB | 4064 | brain01, 2026-11-03 |

### context-free twins online, Java port (JMC) against the Rust implementation's export: pair **asyn-fault**, model fifo, expected violation

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 36 | | conforms* | 6.4 | 156 | 1 h, 250 GB | | brain01, 2026-11-03 |
| 3 | 280 | 400 | 32 | violation* | 9.1 | 183 | 1 h, 250 GB | 4057 | brain01, 2026-11-03 |
| 4 | 2940 | 4900 | 1218 | violation* | 23.2 | 4255 | 1 h, 250 GB | 5178 | brain01, 2026-11-03 |

**Not run (context-free twins online, Java port (JMC) against the Rust implementation's export).** planned but never scheduled (grid points beyond the ladder): 8 fifo correct, 8 fifo asyn-fault, 9 fifo correct, 9 fifo asyn-fault.

## context-free twins online, Rust implementation (TraceForge) against the Java port's export

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **asyn-fault**, model bag, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 144 | 144 | conforms | 3.6 | 157 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 14400 | 14400 | conforms | 26.7 | 5907 | 1 h, 250 GB | | brain01, 2026-11-02 |

### context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **asyn-fault**, model fifo, expected conforms

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 36 | 36 | conforms | 3.4 | 164 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 400 | 400 | conforms | 4.5 | 203 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 4900 | 4900 | conforms | 15.6 | 4079 | 1 h, 250 GB | | brain01, 2026-11-02 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair correct, model bag, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 3.2 | 172 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 280 | 280 | conforms | 4.2 | 204 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 2940 | 2940 | conforms | 13.9 | 3634 | 1 h, 250 GB | | brain01, 2026-11-02 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair correct, model fifo, expected conforms**

| size | $G_{spec}$ | $G_{impl}$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 30 | conforms | 3.1 | 156 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 280 | 280 | conforms | 4.2 | 210 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 4 | 2940 | 2940 | conforms | 14.7 | 3647 | 1 h, 250 GB | | brain01, 2026-11-02 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair asyn-fault, model bag, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 144 | 12 | violation | 3.2 | 172 | 1 h, 250 GB | 2 | brain01, 2026-11-02 |
| 3 | 280 | 14400 | 3438 | violation | 4.7 | 204 | 1 h, 250 GB | 4 | brain01, 2026-11-02 |

**context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair asyn-fault, model fifo, expected violation**

| size | $G_{spec}$ | $G_{impl}$ | $G_x$ | verdict | T | M | tier | first miss ms | run |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 30 | 36 | | conforms* | 3.1 | 156 | 1 h, 250 GB | | brain01, 2026-11-02 |
| 3 | 280 | 400 | 32 | violation* | 4.3 | 210 | 1 h, 250 GB | 3 | brain01, 2026-11-02 |
| 4 | 2940 | 4900 | 1218 | violation* | 14.8 | 3647 | 1 h, 250 GB | 3 | brain01, 2026-11-02 |

**Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export).**
planned but never scheduled (grid points beyond the ladder): 8 fifo correct, 8 fifo asyn-fault, 9 fifo correct, 9 fifo asyn-fault.
skipped: a smaller size had reached its limit: 8 fifo asyn-fault.